

Travel-Time Tomography: Inversion of First-Arrival Data to Retrieve Velocity-Depth Models

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Problem Statement

Goal: Estimate subsurface velocity model $v(\mathbf{x})$ (or slowness $s(\mathbf{x}) = 1/v(\mathbf{x})$) from first-arrival travel-time picks d_j

Applications:

- Near-surface surveys (bedrock mapping, groundwater)
- Deep crustal imaging
- Exploration seismology

Continuous Forward Problem

In ray-theory (high frequency) limit, travel time between source $\mathbf{x}_s$ and receiver $\mathbf{x}_r$:

$$T(\mathbf{x}_s, \mathbf{x}_r; s) = \int_{\Gamma(\mathbf{x}_s, \mathbf{x}_r)} s(\mathbf{x}) d\ell \quad (1)$$

where:

- $s(\mathbf{x}) = 1/v(\mathbf{x})$ is slowness
- Γ denotes ray path (Fermat's principle: shortest travel time)

Discretization

Partition domain into N_m cells with piecewise constant slowness:

$$T_j(\mathbf{s}) = \sum_{i=1}^{N_m} L_{ji} s_i \quad (2)$$

where L_{ji} = ray length in cell i for data j

Jacobian (Sensitivity):

$$J_{ji} = \frac{\partial T_j}{\partial s_i} = L_{ji} \quad (3)$$

Each row = ray path lengths; each column = sensitivity to one cell

Inverse Problem Formulation

Regularized nonlinear least-squares:

$$\Phi(\mathbf{m}) = \|W_d(\mathcal{F}(\mathbf{m}) - \mathbf{d})\|_2^2 + \lambda \|W_m(\mathbf{m} - \mathbf{m}_0)\|_2^2 \quad (4)$$

- W_d = data weighting
- W_m = regularization operator (smoothness)
- λ = regularization parameter
- $\mathbf{m}_0$ = reference model

Gauss-Newton Inversion

Linearize around current iterate $\mathbf{m}^k$:

$$\mathcal{F}(\mathbf{m}^k + \Delta\mathbf{m}) \approx \mathcal{F}(\mathbf{m}^k) + J^k \Delta\mathbf{m} \quad (5)$$

Normal equations:

$$((J^k)^T W_d^T W_d J^k + \lambda W_m^T W_m) \Delta\mathbf{m} = \text{RHS} \quad (6)$$

Update: $\mathbf{m}^{k+1} = \mathbf{m}^k + \Delta\mathbf{m}$

Iterate until convergence (χ^2 target, small updates, or max iterations)

Dijkstra Ray Tracing

Solves eikonal equation numerically:

$$|\nabla T(\mathbf{r})|^2 = \frac{1}{v^2(\mathbf{r})} \quad (7)$$

Algorithm:

- 1 Initialize $T(\text{source}) = 0$, all others $= \infty$
- 2 Build graph: nodes = mesh vertices, edges = connections
- 3 Edge weight = distance / v_{avg} (slowness-weighted)
- 4 Dijkstra's shortest-path $\rightarrow$ minimum travel times

Complexity: $O(n \log n)$ where n = number of mesh nodes

Location: Alpine region near Königsee Lake

Configuration:

- 63 sensors (shots + geophones) along surface line
- 714 shot-receiver pairs
- Travel-time range: 0.35–30 ms
- SGT format: {shot, geophone, time, valid}

Apparent velocity:

$$v_{\text{app}} = \frac{\Delta x}{\Delta t} = \frac{|x_g - x_s|}{t(x_s, x_g)} \quad (8)$$

Koenigsee: Travel-Time Curves

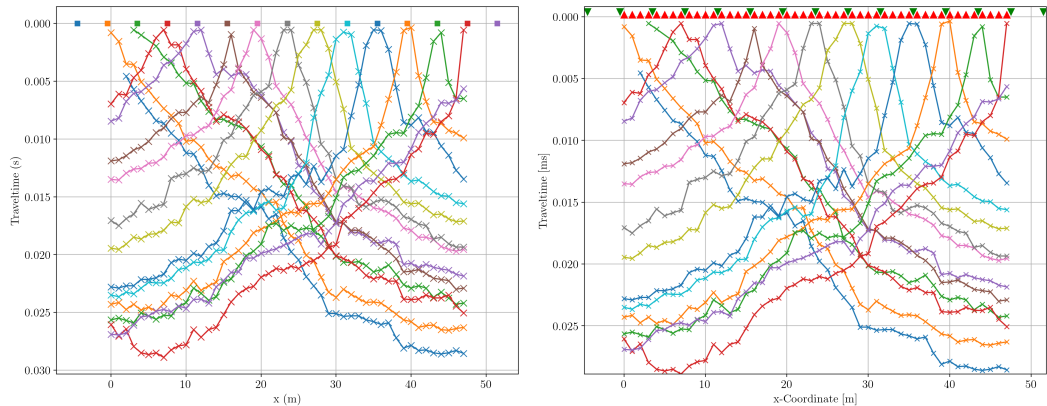


Figure: First-arrival travel-time curves. Each curve = different shot position. Green triangles = shots, red pyramids = geophones.

Koenigsee: Apparent Velocity Matrix

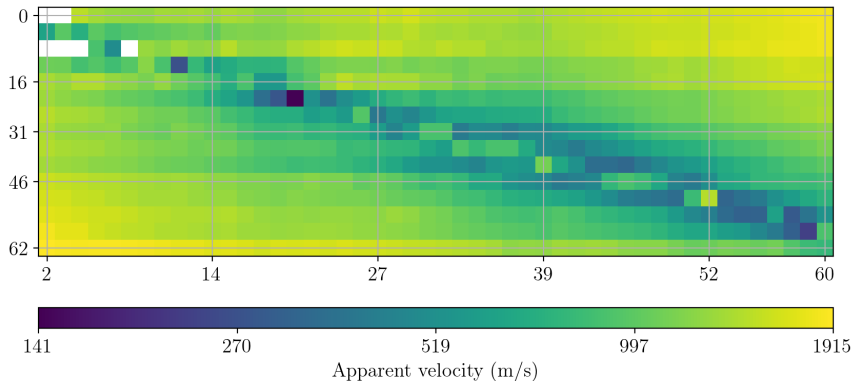


Figure: Apparent velocity matrix. Shot positions (horizontal), geophone positions (vertical). Color = velocity (m/s). Reveals coverage and subsurface heterogeneity.

Koenigsee: Inversion Parameters

Parameter	Value	Description
secNodes	3	Mesh refinement level
paraMaxCellSize	5.0 m	Max cell size
zWeight	0.2	Vertical weighting
vTop	500 m/s	Starting surface velocity
vBottom	5000 m/s	Starting depth velocity
λ	20	Regularization factor

Starting model: $v_{\text{ref}}(z) = v_{\text{top}} + (v_{\text{bottom}} - v_{\text{top}}) \frac{z}{z_{\text{max}}}$

Koenigsee: Inverted Velocity Model

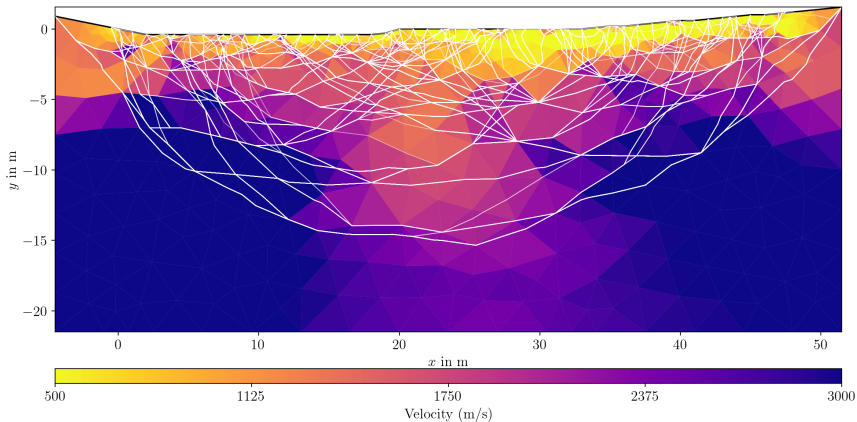


Figure: Final inverted P-wave velocity (m/s). Reveals U-shaped low-velocity zone flanked by higher-velocity regions, consistent with basinal infill above bedrock. Ray coverage shown.

Koenigsee: Predicted Travel Times

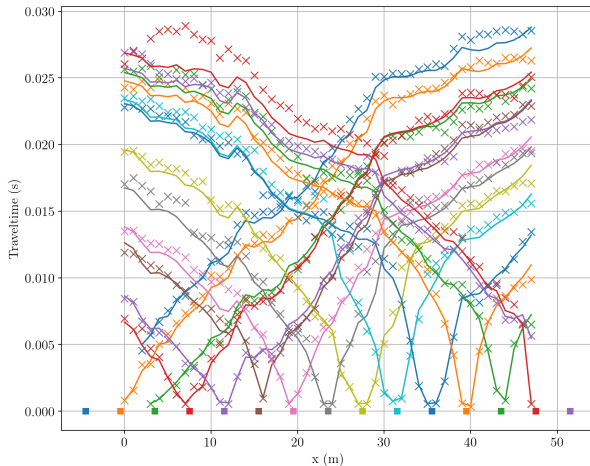


Figure: Predicted travel-time arrivals from inverted model via forward ray tracing. Shows residual misfit.

Synthetic Crosshole Tomography

Objective: Demonstrate inversion in crosshole geometry

Configuration:

- Two vertical boreholes, 20 m apart
- 17 sensors per borehole (1.5 m spacing)
- 34 sensors total
- Domain: $[0, 20] \times [0, -28]$ m
- All shots in Borehole A, all receivers in Borehole B

Rays propagate primarily horizontally (unlike surface refraction)

Crosshole: Survey Geometry

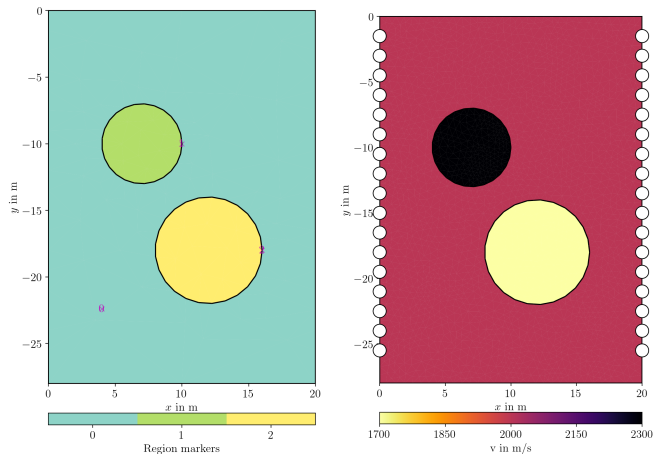


Figure: (Left) Schematic survey geometry. (Right) True velocity model: background 2000 m/s, high-velocity anomaly (2300 m/s) at (7, -10) m, low-velocity anomaly (1700 m/s) at (12, -18) m.

Crosshole: True Velocity Model

Three-component model:

$$v(\mathbf{r}) = \begin{cases} 2000 \text{ m/s} & \text{background} \\ 1700 \text{ m/s} & \text{low-velocity anomaly (R=3 m)} \\ 2300 \text{ m/s} & \text{high-velocity anomaly (R=4 m)} \end{cases} \quad (9)$$

Forward modeling:

- `tt.simulate()` with Dijkstra ray tracing
- Noise: 0.1% relative + 10 μs absolute
- Seed = 1337 (reproducibility)

Crosshole: Apparent Velocity Matrix

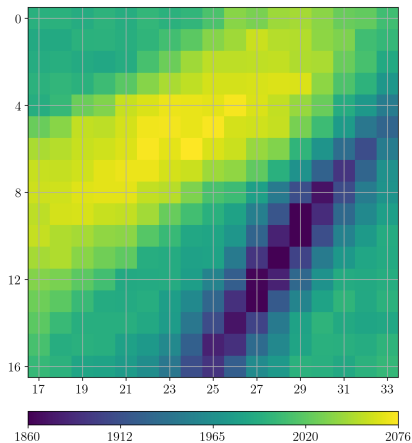


Figure: Apparent velocity matrix plot. Shot positions (horizontal), geophone positions (vertical). Color scale = velocity (m/s). Shows systematic variations.

Crosshole: Inversion Setup

Mesh: Coarse structured grid (0.75 m cells)

- X: [0, 20] m, 28 points
- Z: [0, -28] m, 39 points

Parameters:

- $\text{lam} = 1000$ (strong regularization)
- $\text{zWeight} = 1.0$ (isotropic smoothness)
- $\text{useGradient} = \text{False}$ (homogeneous start: 2000 m/s)
- Model transform: log-slowness
- Convergence: $\chi^2 \leq 1$

Crosshole: Forward and Inversion Meshes

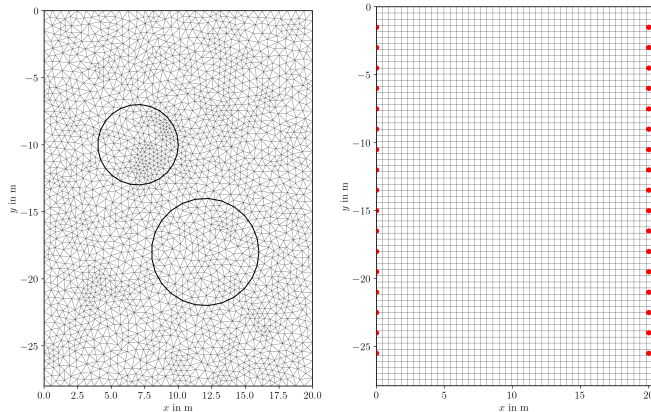


Figure: (Left) Fine triangular mesh for forward modeling. (Right) Coarse structured grid for inversion.

Crosshole: Inversion Results

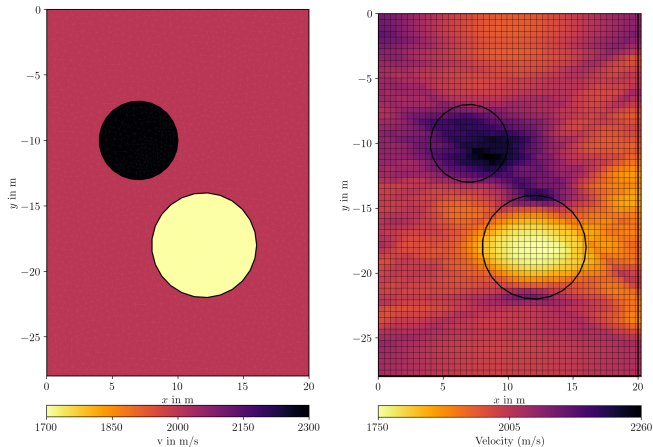


Figure: (Left) True model. (Right) Inverted model. Anomalies recovered with smoothing. Low-velocity (left), high-velocity (right), background (center) correctly identified.

Crosshole: Ray Paths and Coverage

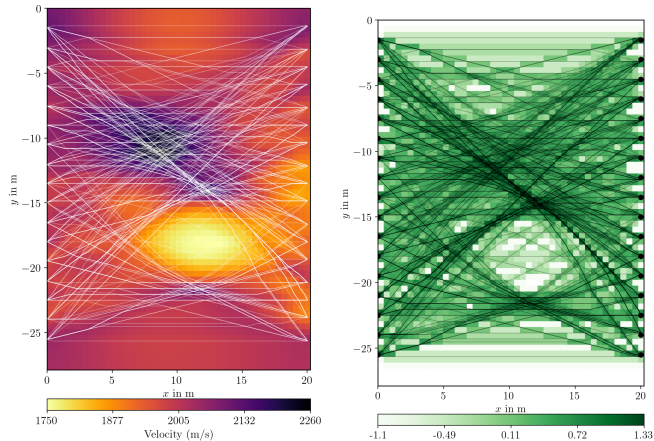


Figure: (Left) Ray paths over recovered model. (Right) Ray coverage map (cumulative lengths). Rays attracted by high-velocity zone, avoid low-velocity region.

Synthetic 2D Profile (Appendix)

Objective: Recover layered velocity from surface refraction

Model: Three sloping layers

- Layer 1: 350 m/s (weathered zone)
- Layer 2: 600 m/s (fractured rock)
- Layer 3: 1500 m/s (bedrock)

Acquisition:

- 48 geophones along sloped surface (117.5 m line)
- Shot spacing: 3 m ($\approx$ 40 shots)
- $\approx$ 1920 ray paths

2D Profile: Geometry

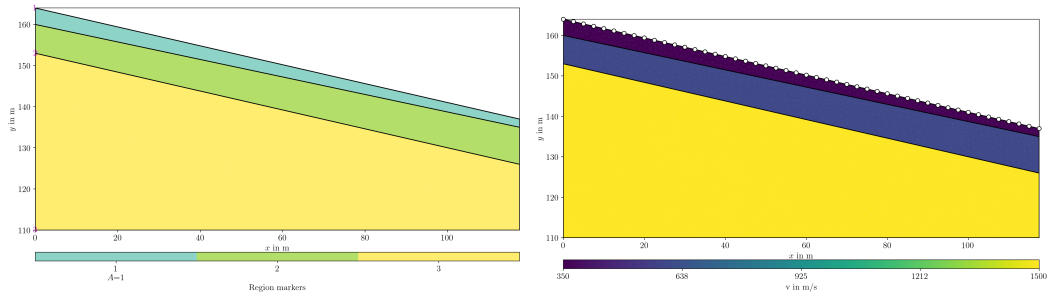


Figure: (Left) Sloping layered geometry. (Right) Geophone and shot positions on surface. White circles = receivers.

2D Profile: Travel-Time Data

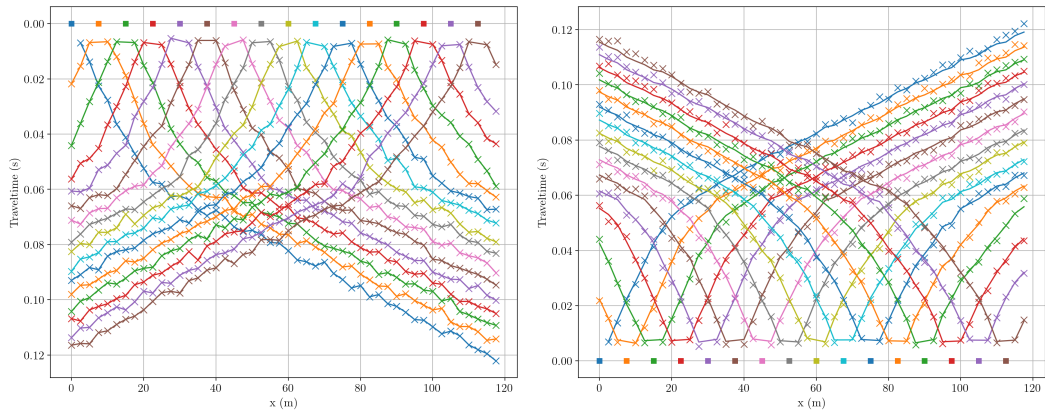


Figure: (Left) Synthetic first-break arrivals. Squares = shots, crosses = geophones. (Right) Predicted arrivals from inverted model.

2D Profile: Inversion Results

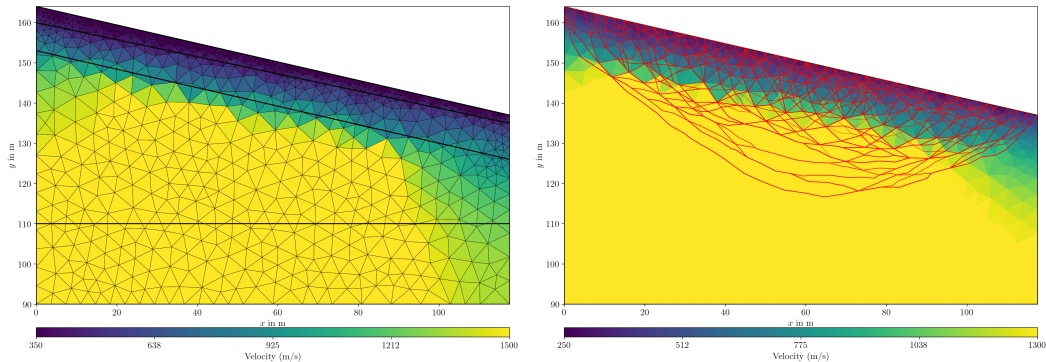


Figure: (Left) Final inverted velocity model. (Right) Ray paths showing illumination of all three layers.

Travel-time tomography via Dijkstra + Gauss-Newton:

- Physically simple (ray paths on mesh)
- Mathematically robust (regularized inverse problem)
- Computationally efficient ($O(n \log n)$ forward solves)

PyGIMLi advantages:

- High-level, user-friendly implementation
- Reproducible workflows (few lines of code)
- Handles field data + synthetic cases

Code: <https://github.com/guntas-13/travel-time-tomography/>

References I



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Thank You!